Kawshik Manikantan

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Education

International Institute of Information Technology (IIITH) Hyderabad, India
B.Tech. + M.S. by Research (dual degree), Computer Science *Jul 2019 – Jan 2025*

- **CGPA:** 9.27/10
- **Awards:** Dean's List (Top 10-15%) in five consecutive semesters (Spring 2021 – Spring 2023).
- **Thesis:** *Coreference without Bells and Whistles*; advisors: Vineet Gandhi and Makarand Tapaswi.
- **Coursework (AI):** Advanced NLP, Computer Vision, Statistical Methods in AI, Advanced Graph Learning, Information Retrieval and Extraction, Digital Image Processing.
- **Coursework (Systems):** Data Structures and Algorithms, Operating Systems and Networks, Design and Analysis of Software Systems.

Publications

MINERVA-Cultural: A Benchmark for Cultural and Multilingual Long Video Reasoning [data]
D. Singh, A. Nagrani, ***K. Manikantan****, H. Singh, D. Tewari, T. Weyand, C. Schmid, A. Angelova, S. Dave* *CVPR 2026 (Highlight)*

IdentifyMe: A Challenging Long-Context Mention Resolution Benchmark for LLMs [code]
K. Manikantan*, M. Tapaswi, V. Gandhi, S. Toshniwal* *NAACL 2025 (Oral)*

Major Entity Identification: A Generalizable Alternative to Coreference Resolution [code]
K. Manikantan*, S. Toshniwal, M. Tapaswi, V. Gandhi* *EMNLP 2024*

Research Experience

Pre-Doctoral Researcher Jul 2025 – Present
Google DeepMind, Languages Group *Bengaluru, India*

- Developing agentic training methods for efficient deep reasoning over long videos.
- Post-training Gemini to improve video reasoning across multilingual and multicultural contexts; co-authored the MINERVA-Cultural benchmark (CVPR 2026, Highlight).
- White-box analysis of cross-lingual factual retrieval in Gemini.

Research Fellow Intern Jan 2025 – Jul 2025
Advanced Micro Devices (AMD), AI Models Group *Bengaluru, India*

- Improved generalization accuracy of a continual model-editing pipeline by ∼15% with multi-faceted data augmentation.
- Built a continually updated, large-scale benchmark of recent Wikipedia edits to measure how well edited models adapt to new facts.

Undergraduate Researcher Apr 2021 – Jan 2025
Center for Visual Information Technology (CVIT), IIITH *Hyderabad, India*

- Proposed Major Entity Identification (MEI), a generalizable alternative to coreference resolution that transfers across domains and enables LLM evaluation (EMNLP 2024).
- Built IdentifyMe, a long-context mention-resolution benchmark for evaluating frontier LLMs (NAACL 2025, Oral).
- Multi-view detection and tracking in long sports videos, in collaboration with Star Sports.

Research Intern Jul 2022 – Sep 2022

Star Sports *Remote, India*

- Built and deployed an operator-free player-tracking system used live during the Asia Cup 2022 broadcasts.
- Drove ideation and feature development, and built the pre-deployment pipeline.

Teaching and Service

Teaching Assistant, Statistical Methods in AI, IIITH (Fall 2023): designed assignments, graded coursework, and conducted tutorials.

Reviewer: 7th Workshop on Narrative Understanding (WNU), NAACL 2025.

Mentoring and organizing: mentored incoming first-year students at IIITH; organizing team of Felicity, the institute's techno-cultural fest.

Selected Projects

A Study of Tracking | *PyTorch, PyTorch Geometric* *github*

- Re-implemented *Tracking at 100 FPS* (GOTURN) and the *neural MOT solver* (MPNTrack), and extended the latter towards multi-view tracking.

Modeling for NLP Tasks | *PyTorch Lightning, Keras, NLTK* *NLI*, *CREPE*, *Mental health*

- Compared Bag-of-Words, LSTM, and Densely Interactive Inference Network (DIIN) models for natural language inference.
- Implemented character-level convolutional (CREPE) and attention-based (CANINE) classifiers and benchmarked them on text classification.
- Multi-class classification of mental-health disorders from Reddit posts, comparing CNN, Random Forest, SVM, XGBoost, and DistilBERT classifiers.

SimpleRA DBMS | *C++* *github*

- Relational DBMS with integer tables and matrix operations: two-phase on-disk merge sort, delayed-write buffer manager, linear-hash and B+-tree indexes.

Wikipedia Search Engine | *Python, NLTK* *github*

- Search engine over a full Wikipedia dump, built from scratch: field-aware inverted index (title, infobox, body, categories, links, references) built by external k-way merging of partial indexes, with stemming, stop-word filtering, and field-restricted queries.

Linux Mini Shell | *C* *github*

- Shell with piping, redirection, background processes, and common built-ins, implemented directly on POSIX system calls.

Skills

Programming	Python, C/C++, Java, JavaScript, MATLAB
ML and NLP	PyTorch (Lightning, Geometric), TensorFlow/Keras, scikit-learn, XGBoost, Hugging Face Transformers and Datasets, NLTK
Tools	Linux, Git/Mercurial, Bazel